From AI Capability to Organisational Value

A Practical Guide to Realising Productivity and Competitive Advantage

White Paper

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Executive Summary

Artificial intelligence is becoming more accessible. Powerful models, cloud based services, and AI enabled tools are now available to organisations of different sizes and sectors. At the same time, governments and business leaders increasingly view AI as a driver of productivity, competitiveness, and innovation.[1]

Yet access to technology alone does not create value. As AI applications diffuse across organisations and sectors, the challenge shifts from access to effective application. Achieving tangible outcomes requires organisations to identify where AI can deliver the greatest impact, integrate it into organisational processes, and develop the capabilities required to realise and scale its benefits.

This paper argues that the central challenge is no longer access to AI technologies, but identifying where AI can contribute most effectively and creating the conditions required to achieve and sustain desired outcomes.

To address this challenge, the paper proposes the AI Capability-to-Value Framework, which links problem identification, value opportunity, AI fit, organisational capacity, deployment, scale, and value creation. The framework provides organisations and policymakers with a practical way to assess where AI can contribute to productivity, competitiveness, and performance, and what is required to realise these outcomes.

As AI becomes a general-purpose technology, competitive advantage depends less on access and more on effective application. Organisations most likely to benefit from AI will not necessarily be those that adopt it most aggressively, but those that identify the right opportunities, deploy AI where it can contribute most effectively, and develop the conditions required to achieve and sustain its benefits.

For executives and policymakers, the key question is no longer:

> How can AI be adopted?

It is:

> Where can AI create the greatest value, and what is required to realise its benefits at scale?

1 From Capability to Value

Advances in AI have significantly expanded the range of tasks and processes that can be automated, augmented, or improved. Yet the existence of technical capability does not, by itself, guarantee tangible benefits.

For organisations, access to AI is a starting point rather than an end point. Technology alone does not determine where it should be applied, how success should be assessed, or whether it will generate lasting gains.

Understanding this transition requires moving beyond technology alone and considering the conditions under which desired outcomes are achieved. Section 2 explains why technology adoption and value creation should not be treated as equivalent.

[1] See Stanford HAI (2025) for evidence on increasing AI adoption, investment, deployment, and economic significance across sectors.

2 Why Adoption Does Not Equal Value

A well-established finding in the economics of innovation is that technology adoption does not automatically translate into economic value.[2]

Technologies often diffuse more rapidly than organisations adapt to them. Productivity gains from major technological advances frequently depend on complementary investments and organisational adjustments that take time to develop. As a result, expectations associated with new technologies often exceed realised performance improvements.[3]

However, recent evidence from workplace applications of generative AI demonstrates that substantial gains can be realised under the right conditions. A study of more than 5,000 customer support agents found significant productivity improvements following the deployment of a generative AI assistant, with larger effects among less experienced workers.[4]

These findings show that AI can generate measurable returns, while also highlighting that outcomes depend on how technologies are implemented within organisations.

The distinction between technological adoption and value creation is particularly relevant in the context of AI. AI may generate benefits through multiple channels, including higher productivity, better decision making, improved service quality, and the reallocation of human effort towards activities requiring greater expertise and judgement. However, these improvements do not arise automatically from access to technology.

In this respect, AI resembles earlier general-purpose technologies. The largest gains are typically realised by organisations that combine technological capabilities with complementary investments.

Technology creates potential, but organisations determine whether and how that potential is translated into meaningful results. Understanding this transition requires a framework that distinguishes between technological possibility and achieved outcomes.

3 The AI Capability-to-Value Framework

If adoption does not automatically create value, organisations require a way to distinguish between technological possibility and realised outcomes.

The objective is not simply to determine whether AI can be deployed. It is to assess where AI can contribute most effectively, whether it is the appropriate solution, and what is required to realise and sustain its impact.

The AI Capability-to-Value Framework offers a structured way to address these questions.

The framework is intended as a practical guide for assessment and decision making. It organises the process of AI value creation into a sequence of interconnected stages, as illustrated below:

It begins with an organisational problem rather than a technological solution.

[2] See, for example, Brynjolfsson, Rock and Syverson (2017); Hall and Khan (2003).
[3] Brynjolfsson, Rock and Syverson (2017).
[4] Brynjolfsson, Li and Raymond (2023).

The sequence reflects the fact that weaknesses in earlier stages can constrain success in later stages. Not every problem represents a significant opportunity. Not every value opportunity requires AI. Not every promising AI application can be deployed or scaled successfully. Each stage therefore provides a basis for assessing how different factors may influence whether value can be realised.

By considering these stages, organisations can better distinguish between technological possibility and achievable value creation. The framework is summarised below.

Problem	A clearly defined operational, strategic, or policy challenge. **Strategic question:** *What problem are we trying to solve?*
Value Opportunity	A plausible source of measurable organisational value. **Strategic question:** *Where can value be created by addressing the problem?*
AI Fit	An assessment of whether AI is suited to the identified problem. **Strategic question:** *Is AI the appropriate solution, or would another intervention be more effective?*
Organisational Capacity	The organisational capabilities and conditions required to realise value. **Strategic question:** *Do we have the capacity required to realise value?*
Deployment	The ability to implement and operate a solution within real organisational processes. **Strategic question:** *Can the solution be successfully deployed in practice?*
Scale	The ability to replicate success across teams, units, functions, or contexts. **Strategic question:** *Can success be replicated at organisational scale?*
Value	The generation of measurable and sustained benefits. **Strategic question:** *Has measurable and sustained value been realised?*

Weaknesses at any stage can limit the ability of organisations to achieve desired outcomes, regardless of the quality of the underlying technology.

A technically successful model may still fail if the problem is poorly defined. A promising pilot may not deliver expected benefits if the organisation lacks the capabilities or conditions required for effective deployment. Similarly, a system that performs well in practice may not scale if responsibilities are unclear, incentives are misaligned, or users lack confidence in its outputs.

The framework highlights that successful AI initiatives depend on more than technological capability alone. The stages provide a structured way to consider factors that influence whether value can be realised and sustained over time.

4 Organisational Capacity as an Enabler of Value

Organisational capacity encompasses the resources, capabilities, and arrangements that shape whether AI initiatives can move from adoption to lasting impact.

Many of the factors associated with successful AI deployment are not technological. Trusted access to relevant data, leadership commitment, workforce capability, governance arrangements, incentives, and process design all influence whether AI can be integrated effectively into organisational activities and scaled over time.

These complementary capabilities help explain why similar technologies often produce different outcomes across organisations.[5] AI may be technically viable, yet fail to generate meaningful improvements if the surrounding organisational environment is not equipped to support its use.

Governance provides a useful illustration of this broader point. It is not an objective in itself, but one component of organisational capacity. Its role is to support the conditions under which AI can be deployed productively, responsibly, and at scale.

This is particularly evident in the context of data. Maximising the benefits of AI depends, in part, on the availability, quality, and reliable use of data. The importance of data as a strategic resource is reflected in policy initiatives such as the European Union's European Strategy for Data, which identifies data as a driver of growth, competitiveness, innovation, and societal progress.[6]

The broader lesson is that technological capability alone is rarely sufficient. Organisations most likely to benefit from AI are those able to identify high-impact opportunities and coordinate technology, people, processes, incentives, and decision-making structures in ways that support effective implementation. Organisational capacity therefore plays an important role in determining whether AI potential is translated into realised outcomes.

5 Productivity and Competitive Advantage

Productivity gains are a central form of value creation, as they reflect the effective conversion of technological capability into measurable improvements in performance. This is closely linked to competitive advantage.

Competitive advantage increasingly depends on an organisation's ability to integrate, adapt, and reconfigure resources in response to technological change.[7] AI capabilities alone are unlikely to confer lasting advantage, as they are becoming widely accessible across sectors. What differentiates organisations is not access to technology, but the ability to combine it with data, skills, organisational processes, managerial judgement, and learning. In this context, advantage depends on how effectively organisations translate AI capabilities into sustained improvements in productivity and performance.

[5] This observation is consistent with broader research emphasising the importance of organisational capabilities in translating technological opportunities into economic and competitive outcomes (Teece, Pisano and Shuen, 1997).

[6] European Commission (2020).

[7] Teece, Pisano and Shuen (1997).

When a technology is scarce, access may itself provide differentiation. As AI tools become more widely available, this source of advantage diminishes. Consistent with the broader literature on competitive advantage, differentiation increasingly depends on how organisations deploy and combine resources rather than on access to technology alone.[8] In the context of AI, this includes selecting the right problems, identifying high-impact opportunities, redesigning workflows, and scaling successful applications.

This implies that AI strategy should be treated less as a technology acquisition exercise and more as a discipline focused on translating potential into results. The central managerial task is not to maximise the number of use cases, but to identify those that justify investment and to establish the conditions required for effective implementation and scale.

For firms, this distinction matters because poorly selected AI initiatives consume resources, divert attention, and create implementation risk without adequate returns. For public institutions, it matters because AI initiatives are intended to improve service quality, administrative efficiency, and policy delivery rather than simply signal technological modernisation.

In both settings, the same principle applies. Competitive advantage and productivity gains do not arise from AI capability alone. They depend on the ability to identify high-impact opportunities, deploy AI effectively, and scale successful applications. The challenge is to translate AI potential into sustained gains.

Conclusion

This paper argues that the central challenge for organisations is not simply the adoption of AI, but identifying where it can generate meaningful impact and developing the conditions required to achieve and sustain desired outcomes.

To address this challenge, the paper proposes the AI Capability-to-Value Framework, linking problem identification, value opportunity, AI fit, organisational capacity, deployment, scale, and value creation. The framework emphasises that successful AI initiatives begin with clearly defined problems rather than technology. It provides a practical way to assess where AI can contribute and what is required to translate that potential into lasting gains.

More broadly, access to AI technologies does not automatically translate into value. AI can deliver substantial improvements, but these depend on complementary investments, effective implementation, and the ability to integrate technology into productive activities.

For organisations, the objective is not to maximise the number of AI applications, but to identify those most likely to deliver meaningful benefits and establish the conditions required for effective implementation and scale. For policymakers, the task is not simply to encourage adoption, but to foster an environment in which these benefits can be realised more widely.

Ultimately, the objective is not simply to expand the use of AI, but to translate technological potential into tangible and sustained outcomes.

[8] See Porter (1996).

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Notes

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